# Chenxi Ma

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## Education

**B.E. in Mechanical Engineering** *Wuhan University of Technology* *09/2024 – Present*

## Project Experience

**High-Performance Inference Framework Development for Ascend NPU** *Tsinghua University* *07/2025 – Present*

- **Framework Adaptation:** Deeply involved in the R&D of an inference framework based on Huawei Ascend hardware; responsible for the low-level adaptation and "Graph Engine (GE)" full-graph offloading for the Qwen 2.5 series.
- **Operator & Graph Optimization:** Optimized computation graph execution paths using Graph Fusion; tuned core operators specifically for the Ascend architecture to improve deployment stability and performance.
- **Heterogeneous Acceleration:** Leveraged Ascend-specific hardware features to achieve high-performance inference and large-scale throughput optimization for Qwen 2.5 on domestic computing platforms.

**Inference Acceleration Research for Mixture-of-Experts (MoE) Models** *Shanghai Jiao Tong University* *02/2025 – 07*

- **Bottleneck Analysis:** Conducted system-level performance profiling focused on expert activation sparsity issues in large-scale MoE models during inference.
- **Optimization Strategy:** Investigated expert activation patterns and explored expert-parallel scheduling; researched overlap of computation and communication to hide latency caused by dynamic routing.
- **Performance Evaluation:** Quantified expert load imbalance across different batch sizes and evaluated execution efficiency on heterogeneous backends to reduce Time-to-First-Token (TTFT) in long-context scenarios.

**Lightweight High-Performance Deep Learning Inference Framework** *Personal Project* *11/2024 – 01/2025*

- **System Design:** Built an extensible inference framework supporting LLaMA 2/3.2, Qwen, YOLOv5, and ResNet with both CPU and GPU backends.
- **Architecture Optimization:** Integrated KV Cache and Continuous Batching; utilized NVIDIA Nsight to optimize CUDA kernels (RMSNorm, FlashAttention, GEMM), achieving a 40% increase in throughput.
- **Quantization:** Implemented INT8 weight-only/group quantization to reduce model footprint and latency while maintaining high precision.

## Professional Skills

- **LLM Inference & MLSys:** Proficient in PagedAttention, KV Cache, and Continuous Batching; skilled in optimizing CUDA operators; experienced with quantization (INT8) and performance profiling tools (Nsight, nvprof).
- **Model Training & Fine-tuning:** Deep understanding of Transformer/BERT/GPT architectures; mastered Parameter-Efficient Fine-Tuning (PEFT) like LoRA; familiar with the full LLM lifecycle (Pre-training, SFT, RLHF).
- **Reinforcement Learning:** Understanding of RL algorithms including Monte Carlo, TD Learning, Q-learning, PPO, and the latest LLM-focused RL methods like GRPO and DPO.
- **Technical Stack:** Proficient in C++ (STL, Templates), CUDA programming, PyTorch, and Huawei CANN/Ascend GE.

## Honors & Awards

**China Undergraduate Mathematical Contest (CMC)** *National Second Prize* *11/2024*